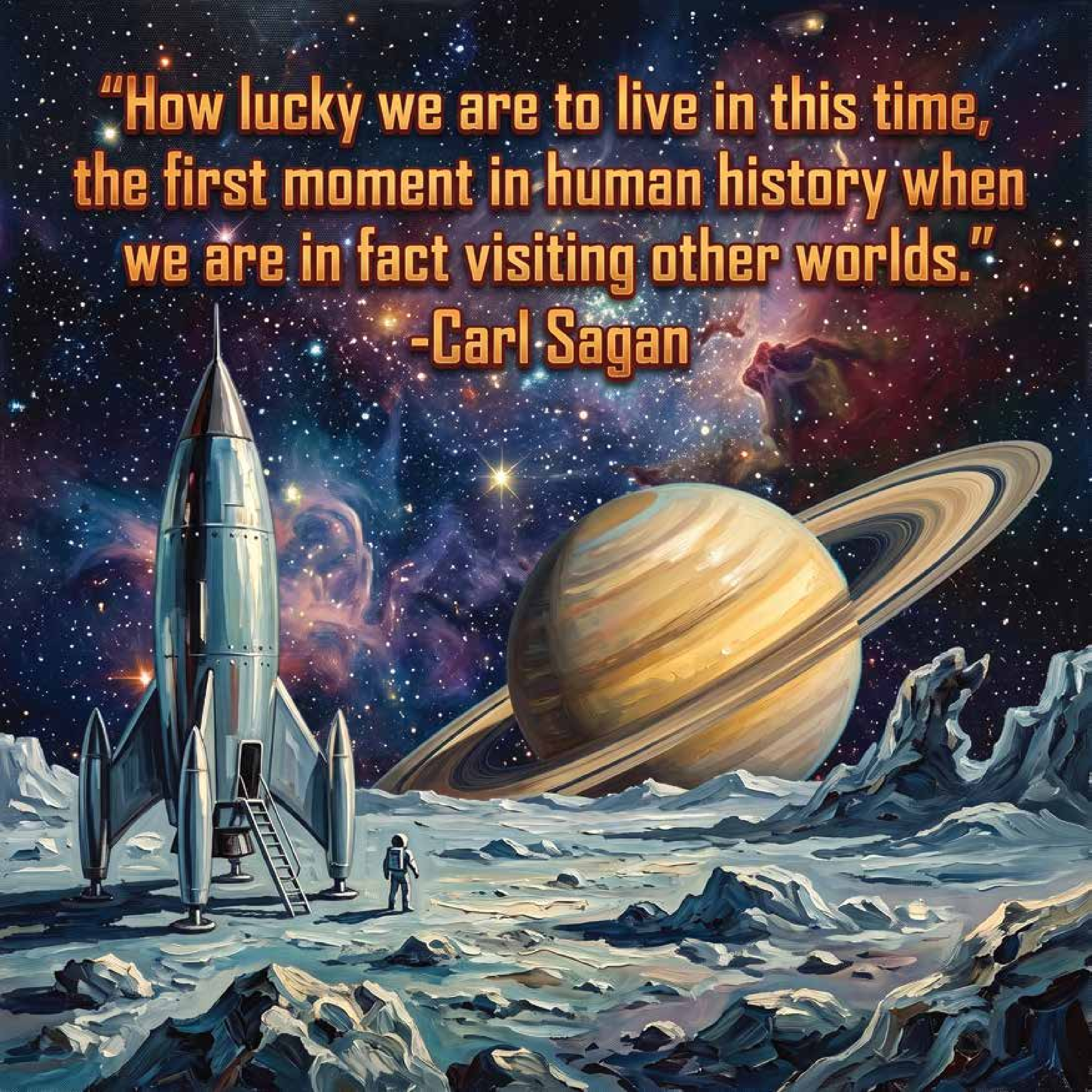
A Brief History of Flight

by Ben Lackey


"How lucky we are to live in this time,
the first moment in human history when
we are in fact visiting other worlds."
-Carl Sagan

Early History

Man has been making flying devices since before written history.

Long before people invented paper airplanes, children threw rocks.

They watched birds fly across the sky and imagined joining them.

They played with maple seeds, contemplating the dynamics of lift.

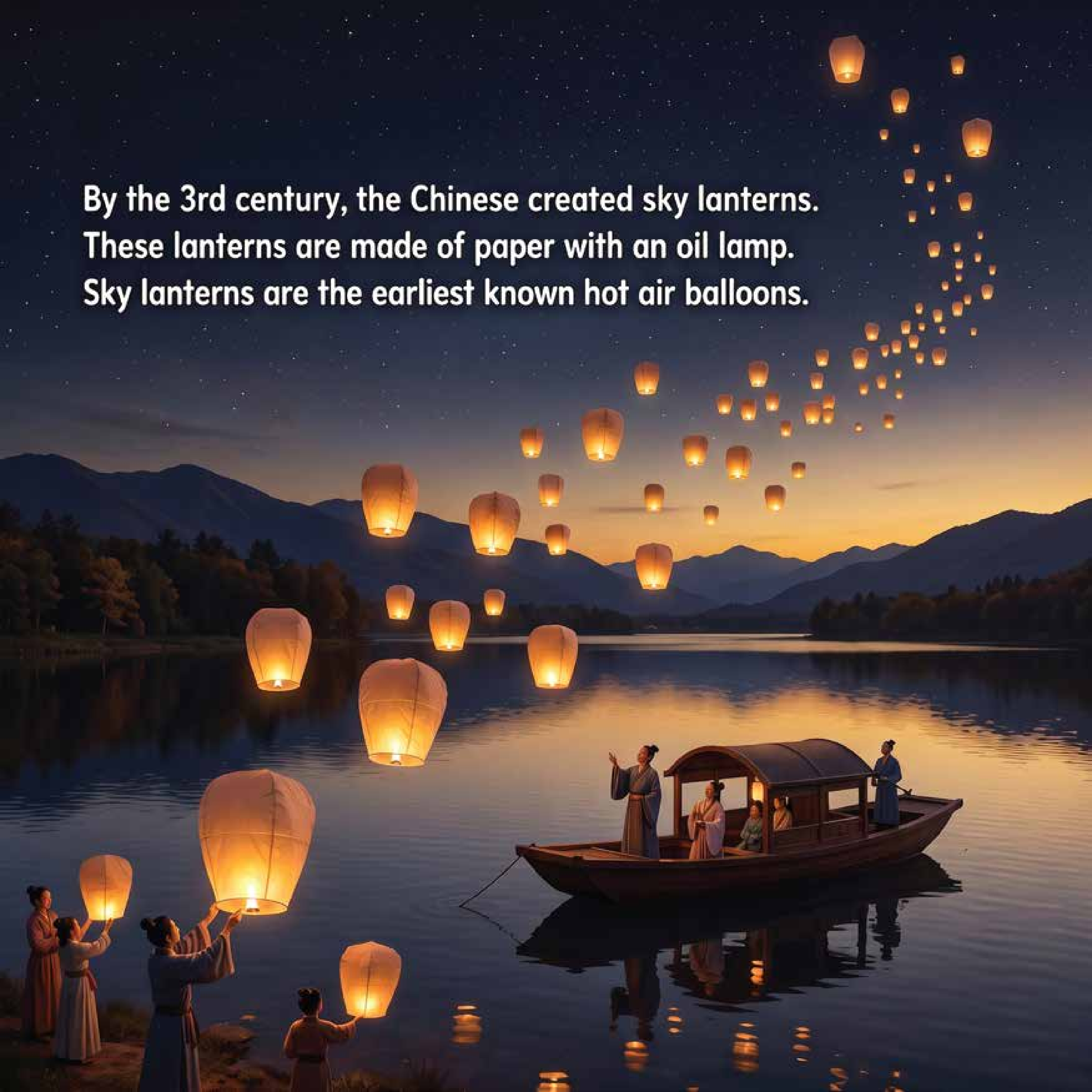
The oldest spears were discovered in South Africa. Those had stone points and are half a million years old. More recent examples appear in Ethiopia, Germany and England.

Early arrowheads date from 64,000 BC. These show cracks consistent with the high velocity of a bow.

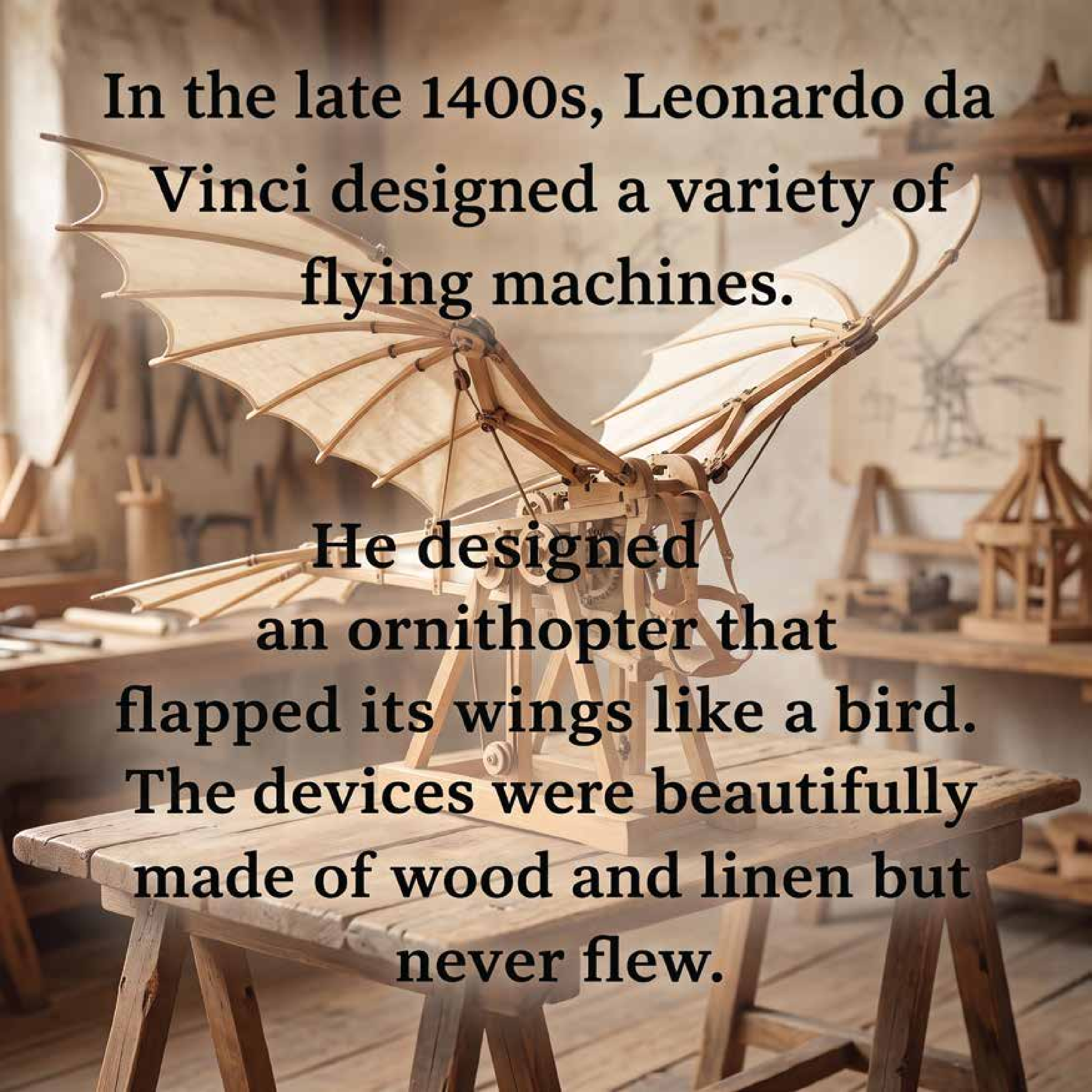
In Poland we've discovered a 30,000 year old mammoth tusk carved into an airfoil.

Across the world in Austrailia, boomerangs have been used for 10,000 years.


Ovid wrote the story of
Icarus. His father builds wings of
feathers and wax. But Icarus flys too close
to the sun. The wings melt and he falls.



By the 3rd century, the Chinese created sky lanterns.
These lanterns are made of paper with an oil lamp.
Sky lanterns are the earliest known hot air balloons.

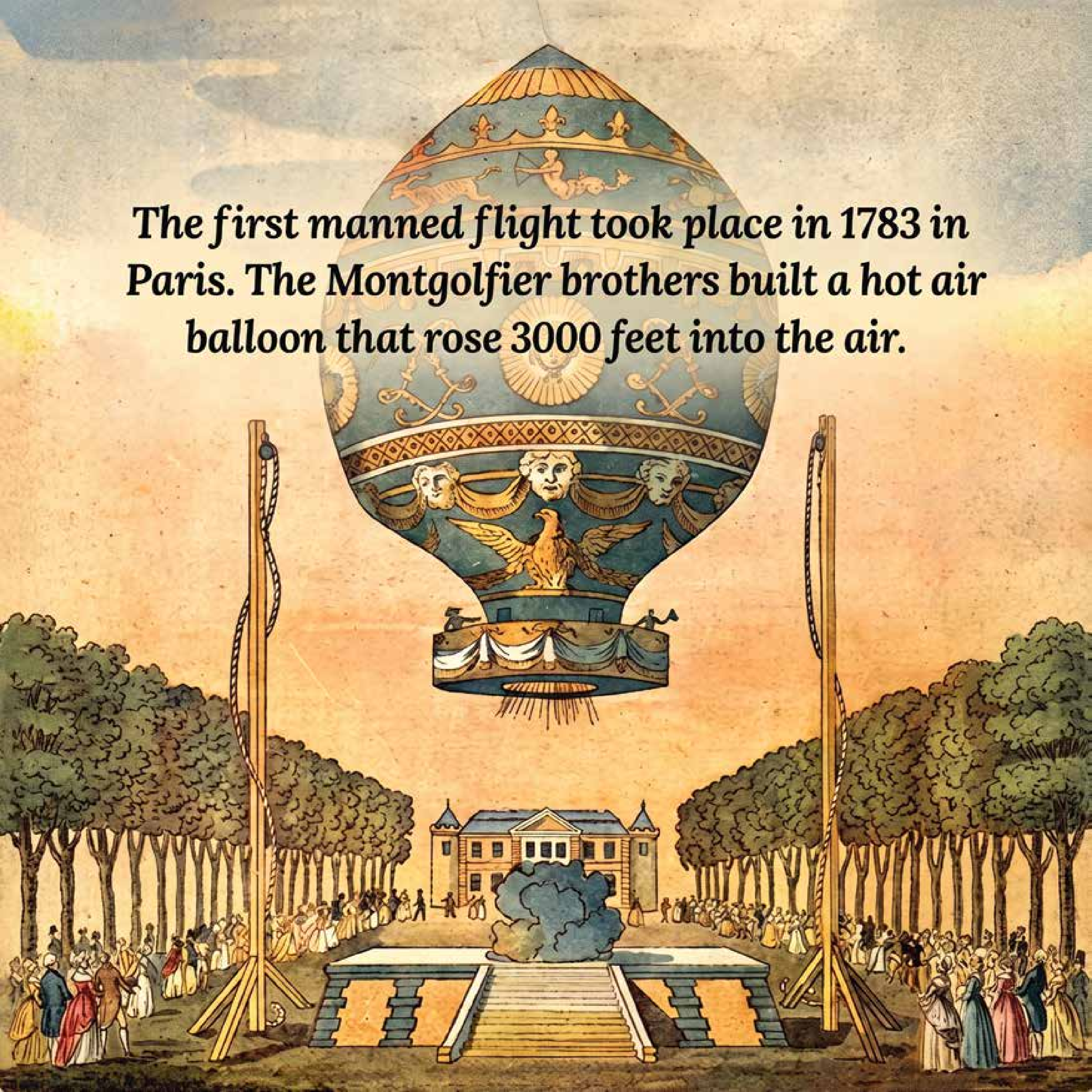


In the late 1400s, Leonardo da Vinci designed a variety of flying machines.

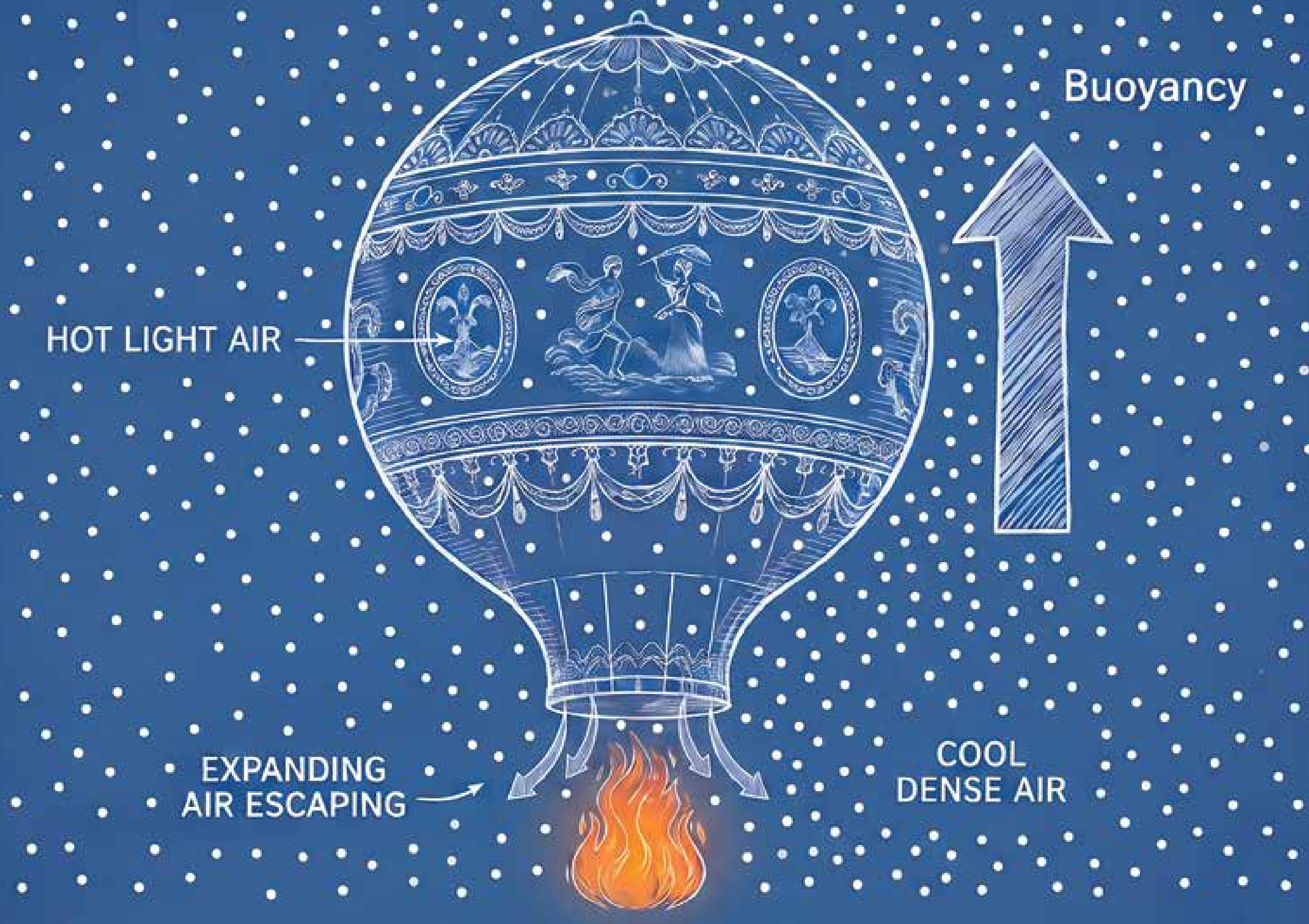
He designed an ornithopter that flapped its wings like a bird. The devices were beautifully made of wood and linen but never flew.


Lighter
than Air

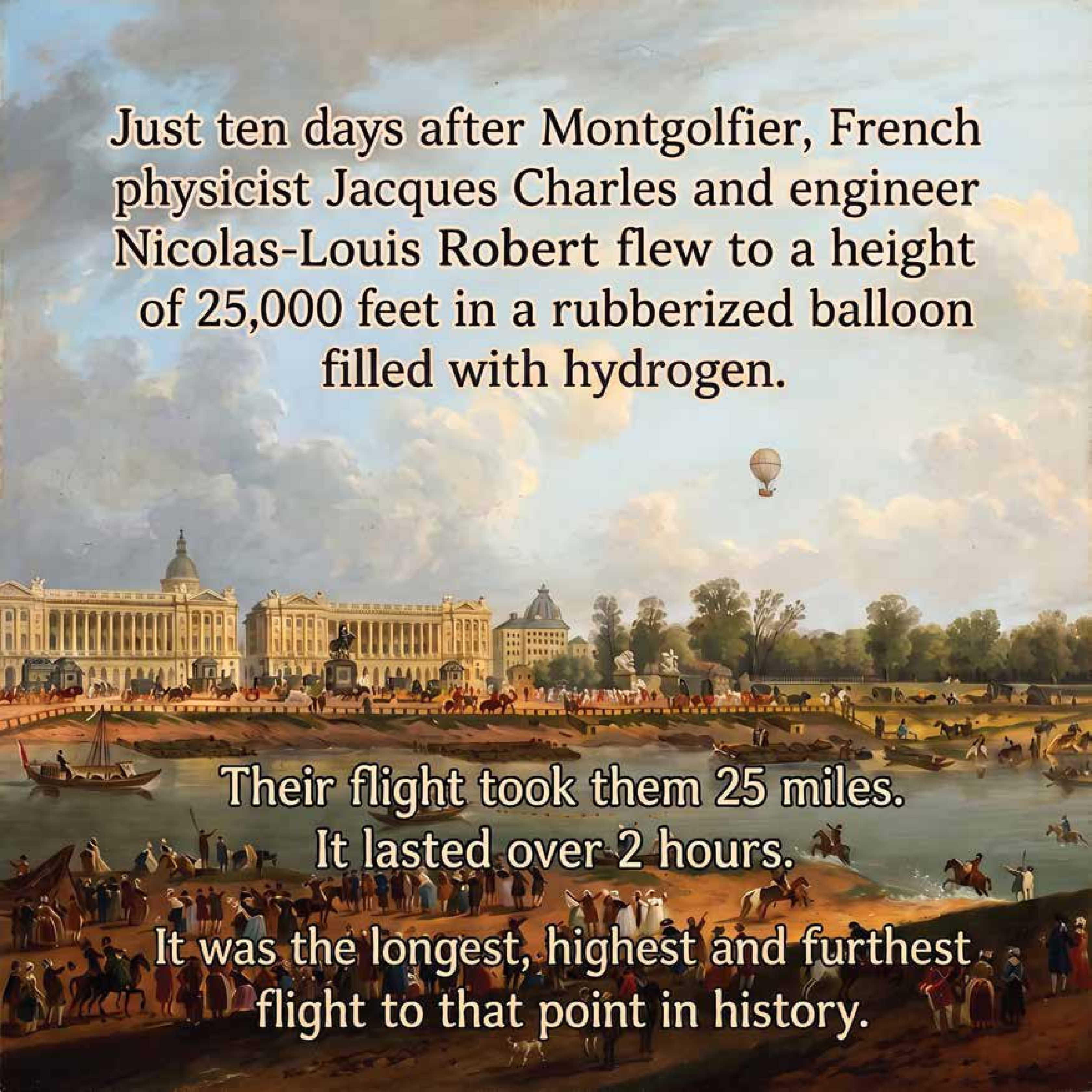
The first manned flight took place in 1783 in Paris. The Montgolfier brothers built a hot air balloon that rose 3000 feet into the air.



A fire heats the air in the balloon.

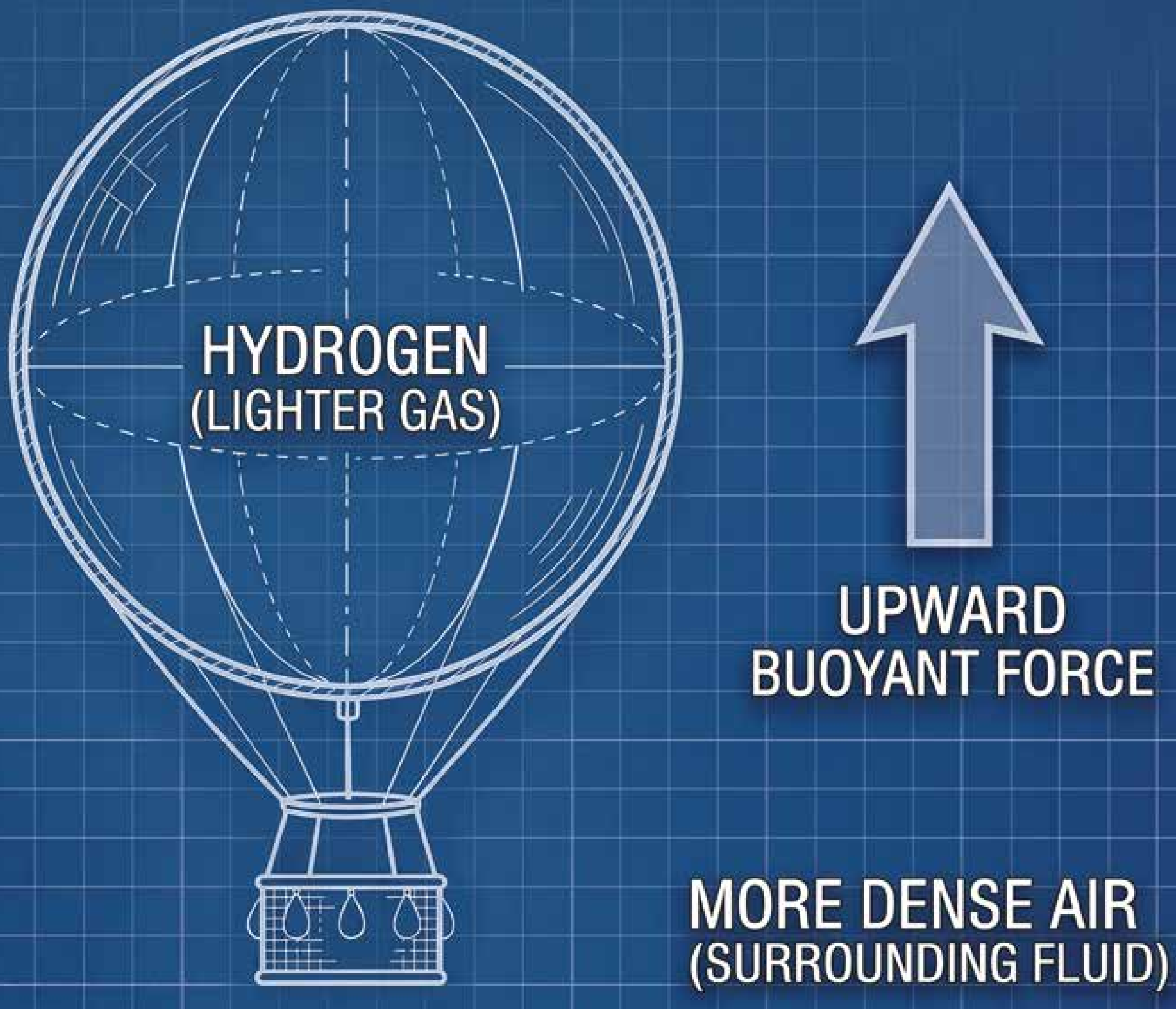


The hot air is less dense than the surrounding air. This causes it to rise.

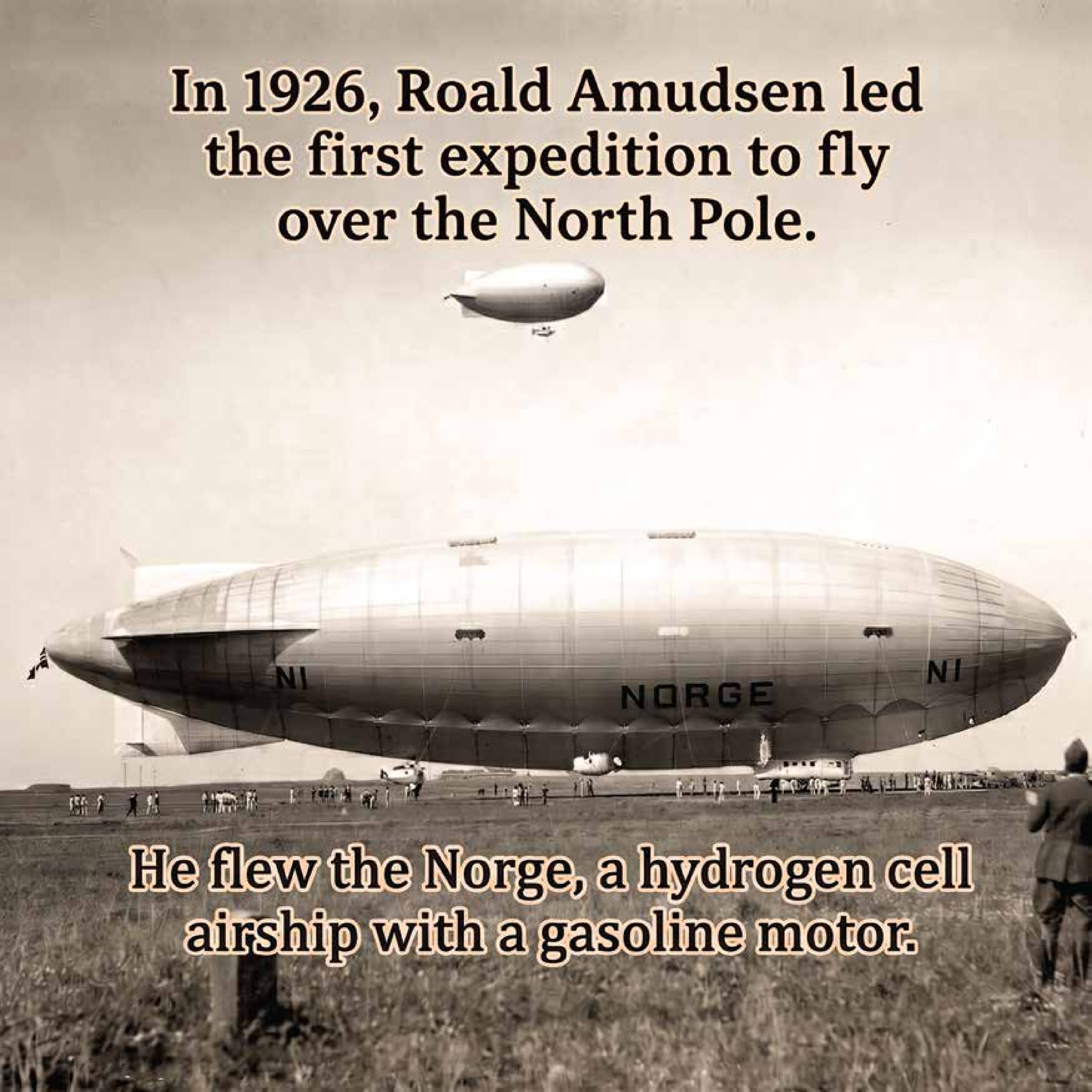


Just ten days after Montgolfier, French physicist Jacques Charles and engineer Nicolas-Louis Robert flew to a height of 25,000 feet in a rubberized balloon filled with hydrogen.
Their flight took them 25 miles.
It lasted over 2 hours.
It was the longest, highest and furthest flight to that point in history.

A LIFTING GAS BALLOON IS FILLED WITH A GAS THAT IS LIGHTER THAN AIR. HYDROGEN AND HELIUM ARE MOST COMMON.



THE BALLOON GENERATES LIFT BASED ON ARCHIMEDES' PRINCIPLE OF BUOYANCY. AN OBJECT IMMERSED IN A FLUID EXPERIENCES AN UPWARD BUOYANT FORCE EQUAL TO THE WEIGHT OF THE FLUID IT DISPLACES.

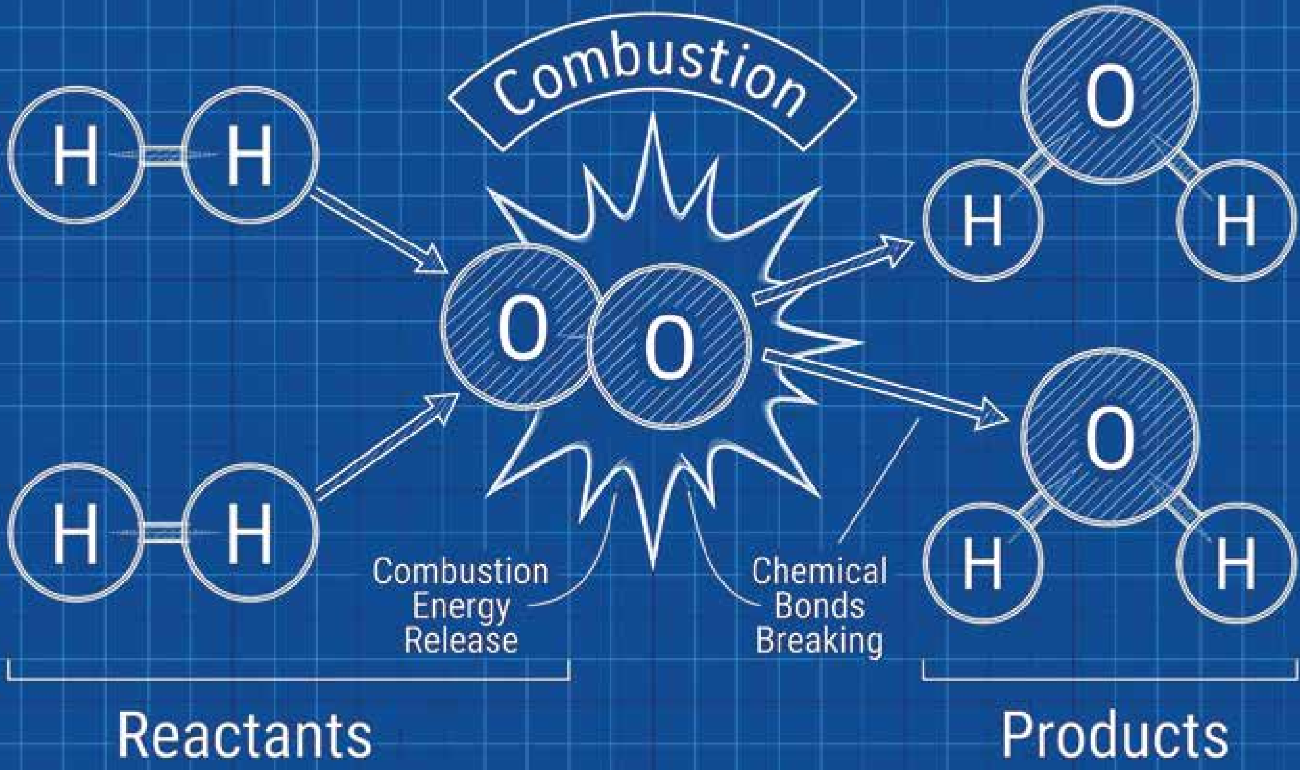

In 1926, Roald Amudsen led the first expedition to fly over the North Pole.
N1
NORGE
N1
He flew the Norge, a hydrogen cell airship with a gasoline motor.

In 1929, the Graf Zeppelin circumnavigated the world, leaving New Jersey and stopping in Germany, Japan, and California.

The ship carried 61 people on a 21-day journey over 20,000 miles.

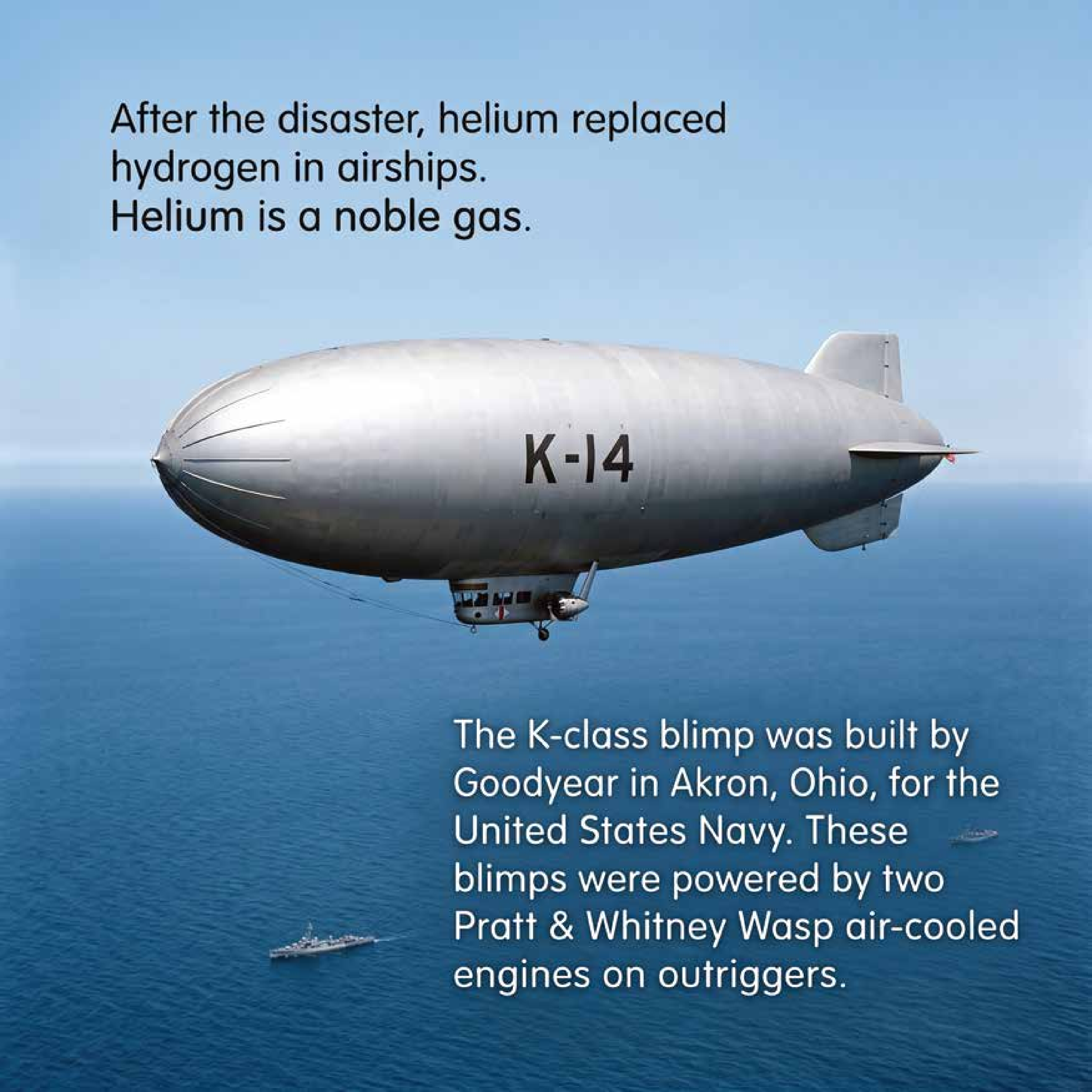

Tragedy struck in 1937.
The Hindenburg caught fire
and crashed.
HINDENBURG
The airship was filled with hydrogen.
Hydrogen is highly flammable.

Hydrogen is extremely reactive. H_2 and O_2 combust to produce H_2O or di-hydrogen monoxide, more commonly known as water.



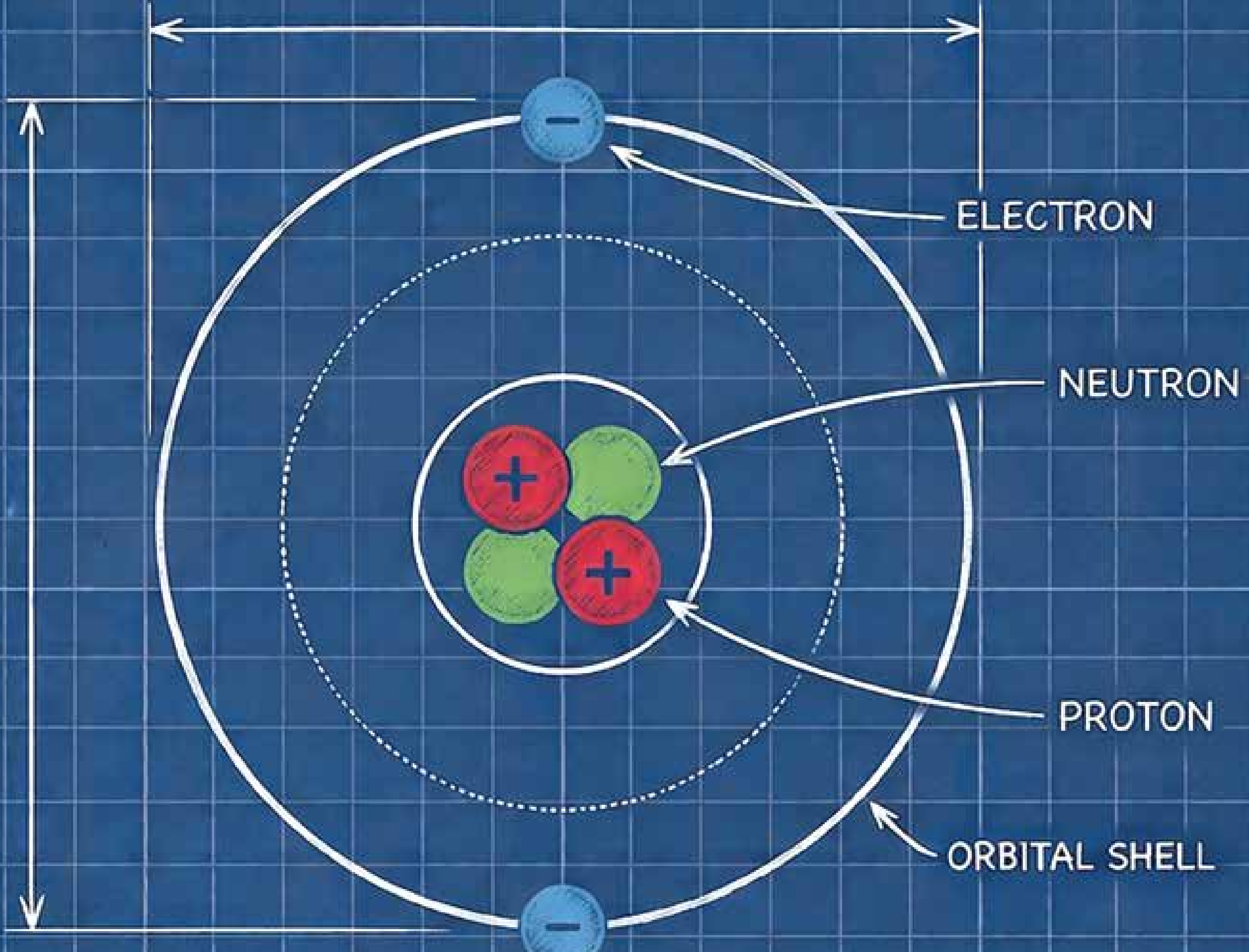
$$2H_2 \text{ (Hydrogen)} + O_2 \text{ (Oxygen)} \longrightarrow 2H_2O \text{ (Water)}$$

After the disaster, helium replaced hydrogen in airships.
Helium is a noble gas.



The K-class blimp was built by Goodyear in Akron, Ohio, for the United States Navy. These blimps were powered by two Pratt & Whitney Wasp air-cooled engines on outriggers.

HELIUM ATOM



Helium cannot form chemical bonds with oxygen to release heat. That is why modern airships use it instead of hydrogen.

Helium is heavier than hydrogen. So, while safer, it is an inferior lifting gas.


In 1960 USAF Captain Joseph Kittinger flew the Excelsior III balloon to 102,800 ft. He then jumped out to test a new parachute.
On the way down he reached a top speed of 614 MPH before deploying his parachute at 18,000ft.

Modern recreational hot air balloons have a propane burner that puts out 10 to 20 million BTU. To contrast a home stove burner might do 20 thousand BTU. Balloons are made of nylon or ripstop polyester. A basket hangs below to carry passengers.

Mazama, near Winthrop, has a hot air balloon festival.


GOODYEAR
Since 1925, Goodyear has operated a fleet of promotional blimps.
The Spirit of Innovation flew from 1986 until 2017. It was 192ft long with a top speed of 50mph. Unlike the Zeppelin, it was a non rigid design.

In 1982, ‘Lawnchair’ Larry Walters attached 42 helium-filled weather balloons to a Sears lawn chair named Inspiration I. He flew to approximately 16,000 feet, drifting into controlled airspace at LAX.

Armed with a pellet gun, he shot several balloons to descend. He dropped the pellet gun and crashed into power lines in Long Beach without serious injury.

On March 21, 1999, Bertrand Piccard and Brian Jones completed the first non-stop, around-the-world flight in a balloon aboard the Breitling Orbiter 3.

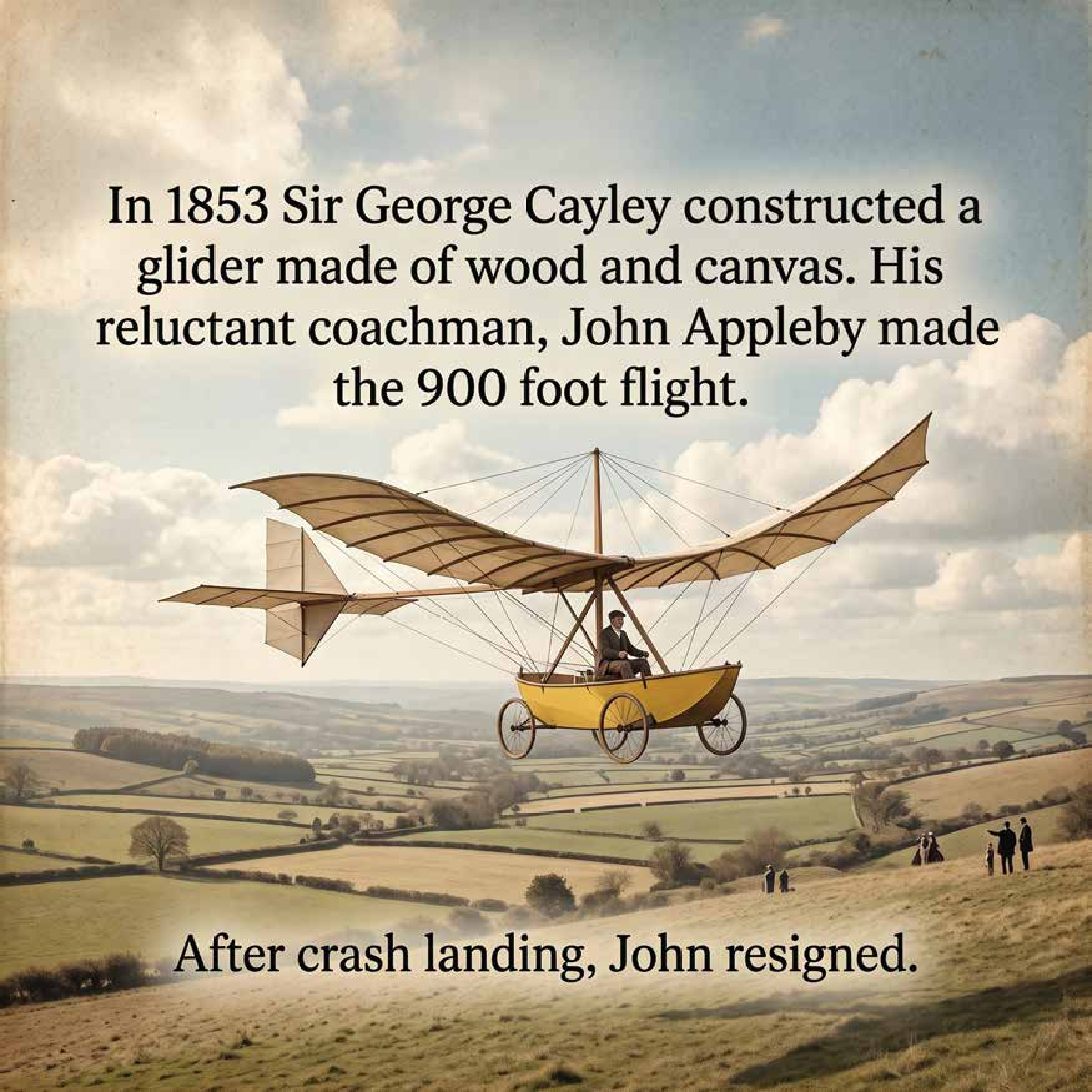
The balloon was a dual gas hybrid. It combined an inner cell filled with helium and an outer envelope heated by propane burners. Solar radiation warmed the helium during the day, while burners heated the air envelope at night.

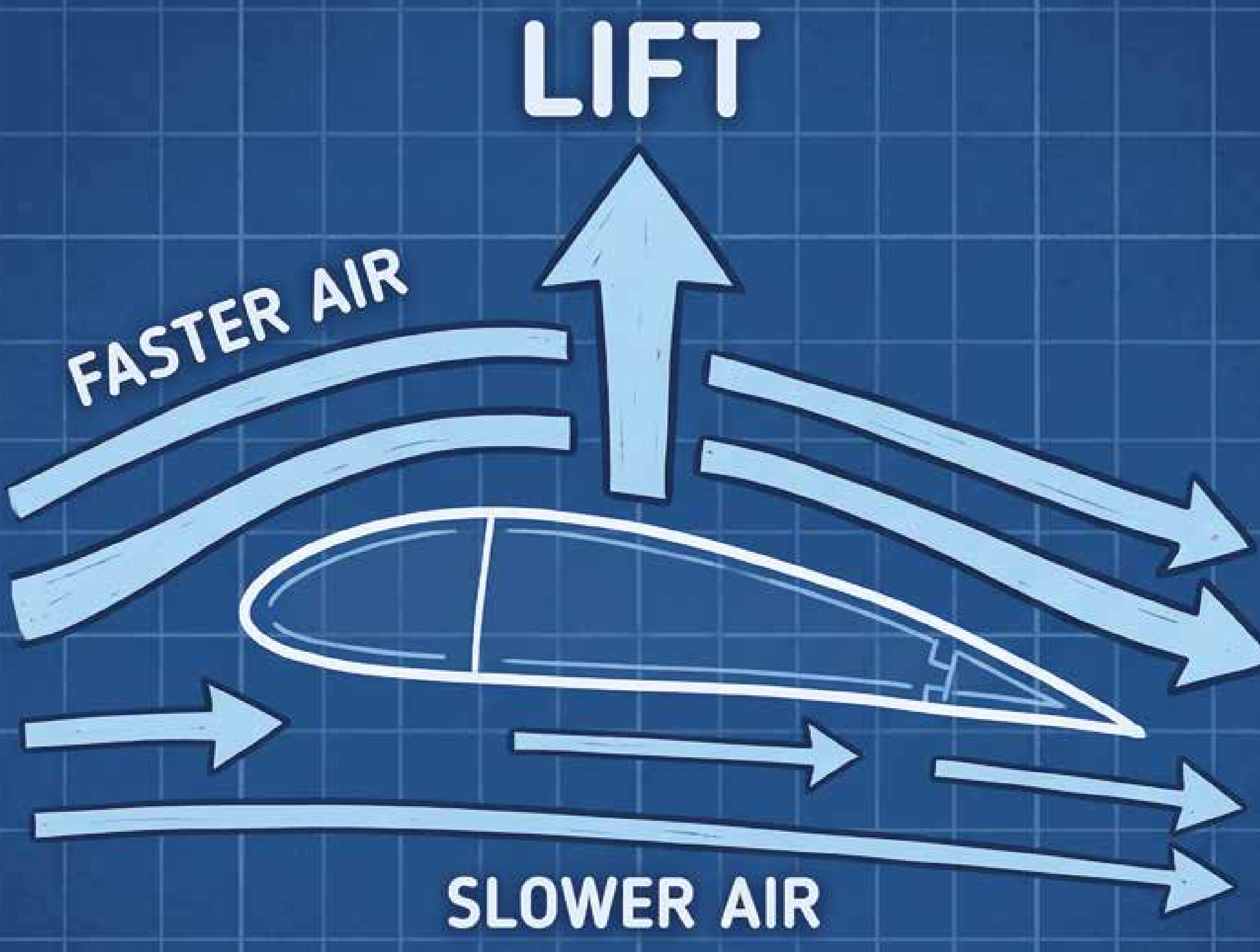
Pathfinder 1 is a rigid airship developed by LTA Research, an aerospace venture founded by Sergey Brin.

Pathfinder 1 measures 406 feet long. It is the first major rigid airship constructed since 1930s. It uses modern material such as carbon fiber and computer controls.


Fixed
Wing
FY A
FY L


FY V
FY B


In 1853 Sir George Cayley constructed a glider made of wood and canvas. His reluctant coachman, John Appleby made the 900 foot flight.
After crash landing, John resigned.

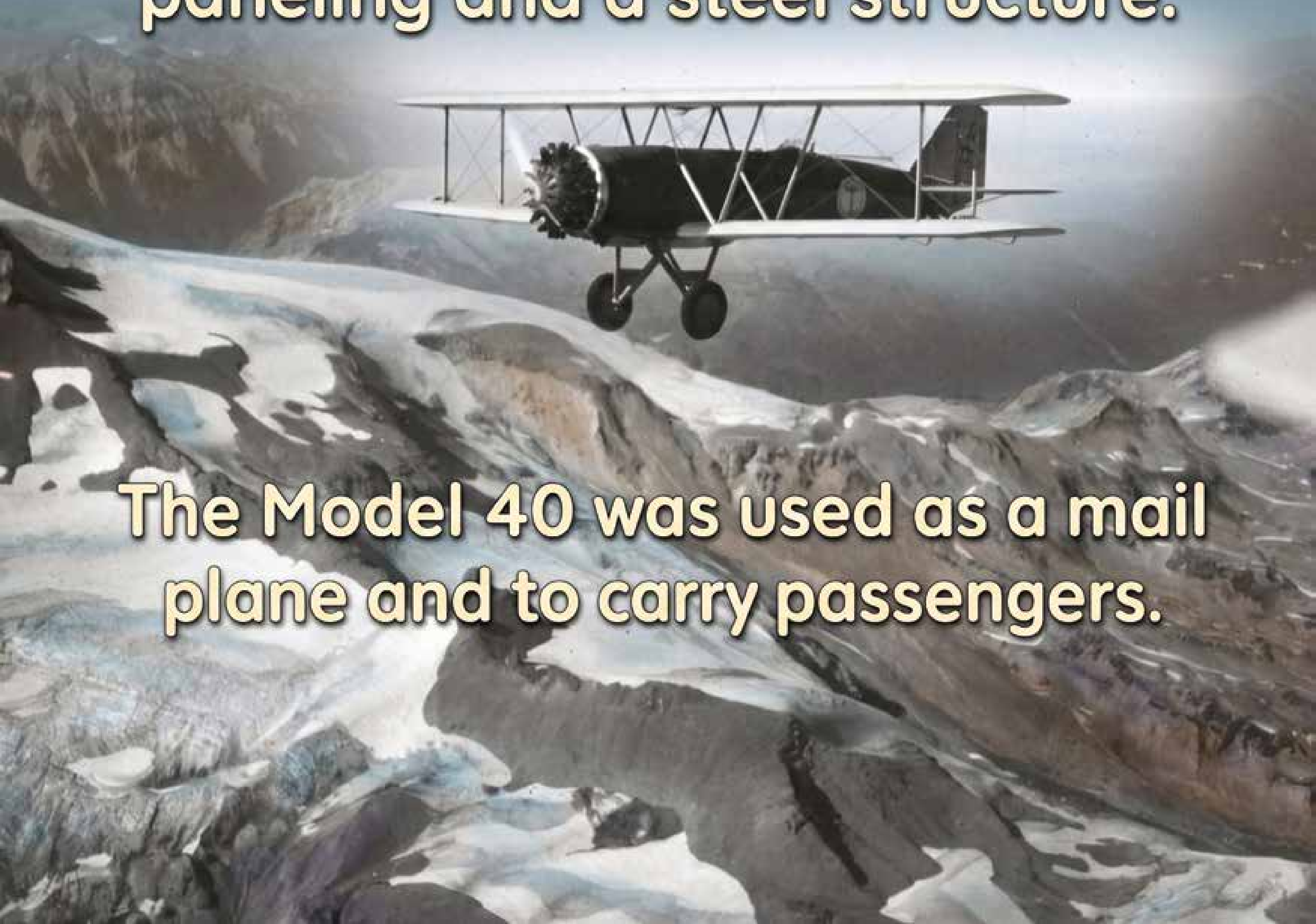


George Cayley established the four fundamental aerodynamic forces of flight: lift, drag, thrust, and weight.

The Wright brothers designed, built and flew the first airplane. In 1903, the Wright Flyer flew 120ft in 12 seconds.

The plane was made of wood, piano wire and muslin wings. A four cylinder gasoline engine provided 4 horsepower.

Planes improved rapidly. By 1925 Boeing introduced the Model 40. This was a biplane with wood paneling and a steel structure.



The Model 40 was used as a mail plane and to carry passengers.

In 1935, Douglas introduced the DC-3.
It is a twin-engine propeller airliner.
It made passenger air service profitable without government subsidies.

The DC-3 was used for transport during World War II under the military designation C-47 Skytrain.

The Boeing B-29 Superfortress was a four-engine, propeller-driven bomber. It had a pressurized cabin, allowing flights up to 31,800 ft.

The Enola Gay and Bockscar dropped atomic bombs on Hiroshima and Nagasaki in 1945. That forced a Japanese surrender, saving tens of thousands of American lives and ending WWII.

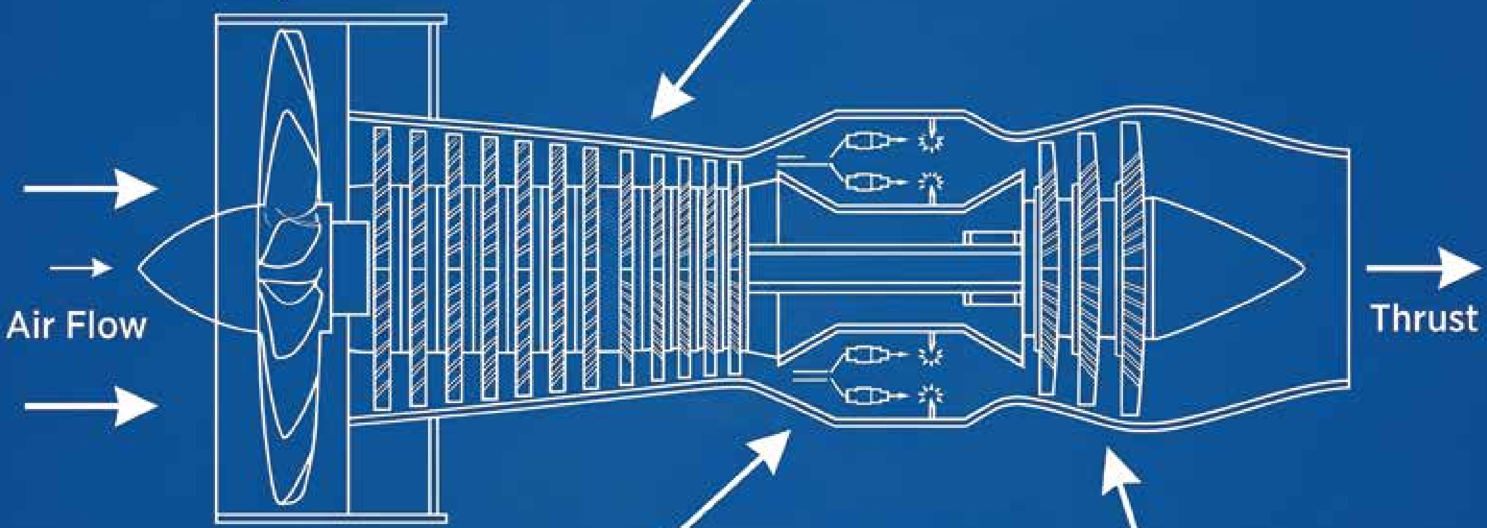
The Messerschmitt Me 262 Schwalbe was the first jet aircraft. Its first powered flight was on July 18, 1942.

The jet outperformed allied planes. However it was complex and expensive. The jet was prone to compressor stalls. Rapid throttle movements at low speeds or high altitudes would often cause flameouts or engine fires.

JET ENGINES HAVE FOUR CONTINUOUS STAGES

1. INTAKE: The fan draws air into the engine.

2. COMPRESSION: Rotating blades tightly compresses the air, raising its pressure and temperature.

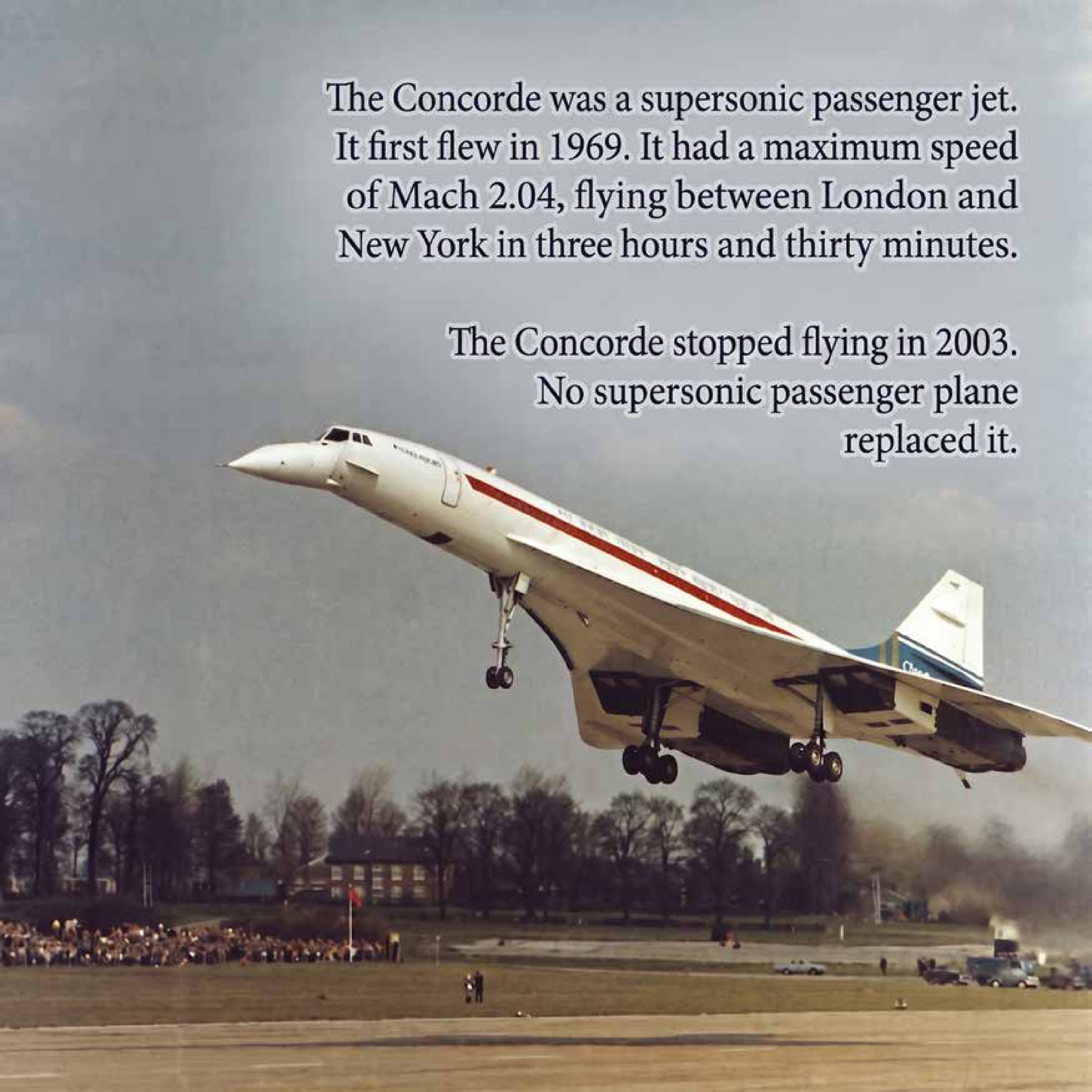


3. COMBUSTION: Pressurized air enters the combustion chamber. Jet fuel is sprayed in and ignited. The burning burning fuel expands, creating heat and pressure.

4. EXHAUST: The expanding gas rushes through turbine blades, spinning them to keep the front fan and compressors running. The gas shoots out through the exhaust nozzle at high velocity, generating the thrust that pushes the aircraft.

The Concorde was a supersonic passenger jet. It first flew in 1969. It had a maximum speed of Mach 2.04, flying between London and New York in three hours and thirty minutes.

The Concorde stopped flying in 2003. No supersonic passenger plane replaced it.

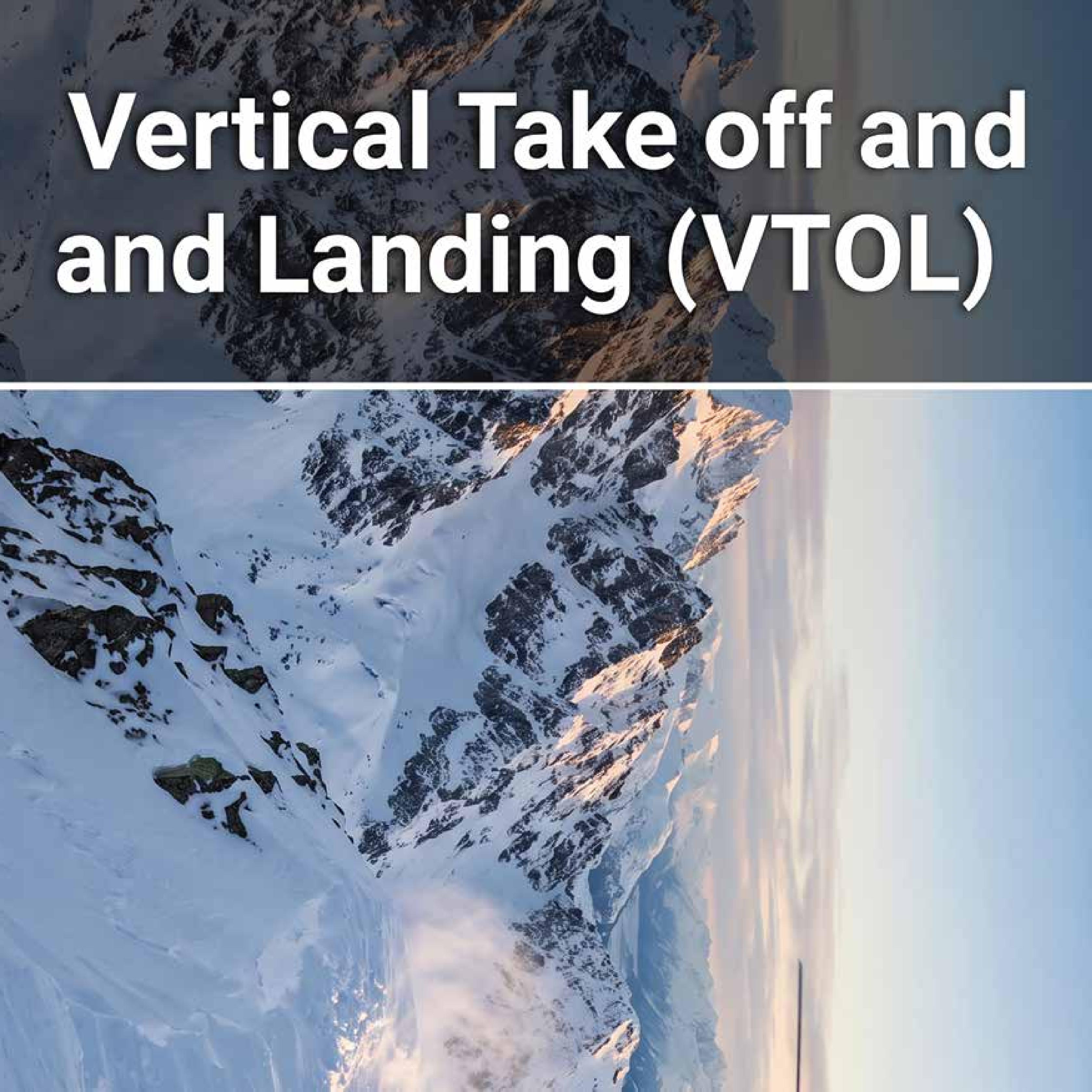


The F-14 Tomcat was introduced in 1974. It had variable geometry wings. These allowed the plane to transform for supersonic flight.


The Rutan Voyager completed the first nonstop, unrefueled flight around the world. Piloted by Dick Rutan and Jeana Yeager, the aircraft flew 26,366 miles across 9 days.
N269VA
The plane is in the the Smithsonian Air and Space Museum. A replica of the Voyager hangs in SeaTac airport.

The B-2 is a flying wing. It doesn't have separate fuselage and wings. This shape reduces its radar signature and allows for a 40,000 lb carrying capacity.

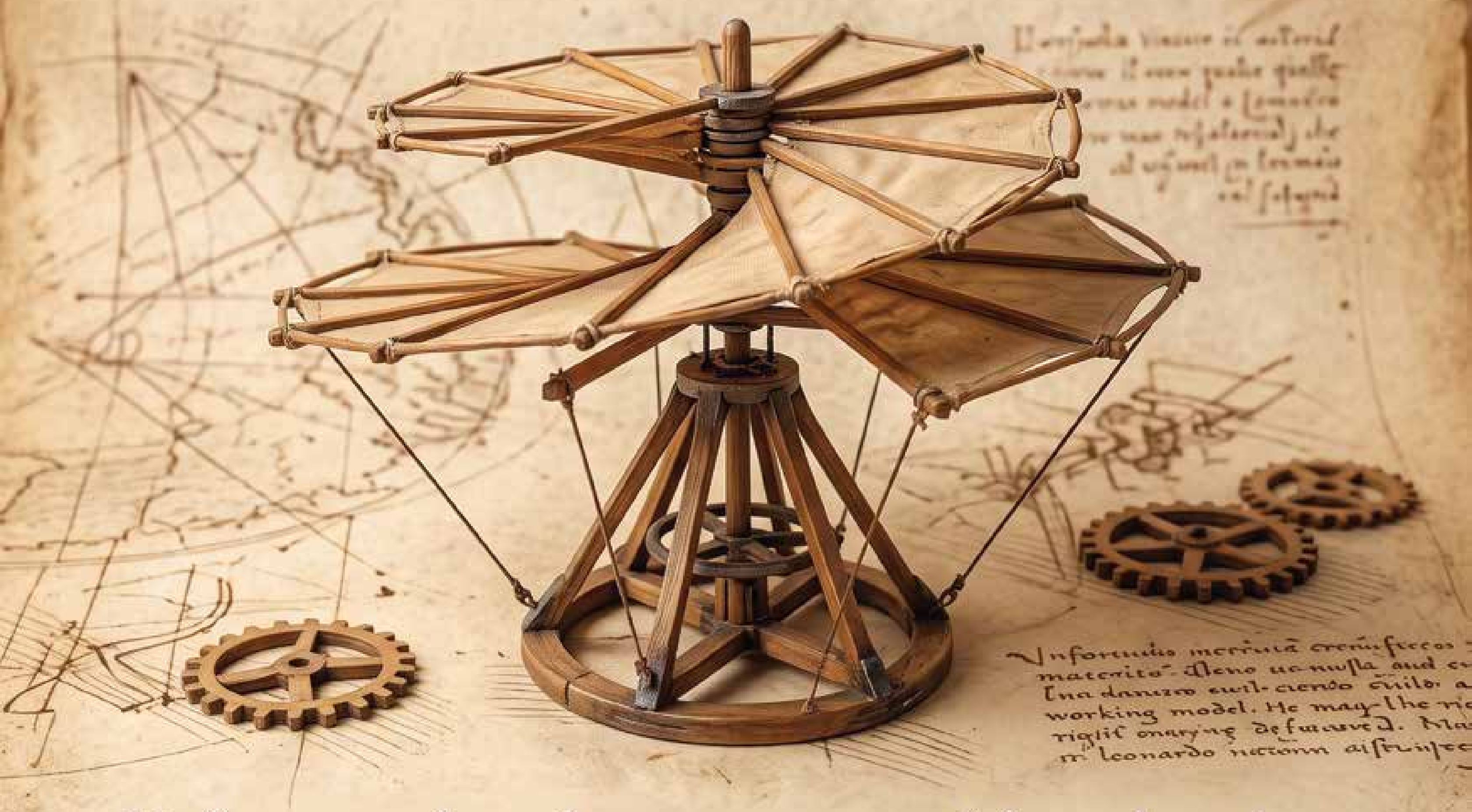
Flying wings may someday replace current civilian aircraft. They have much more interior space, so could make travel more comfortable.



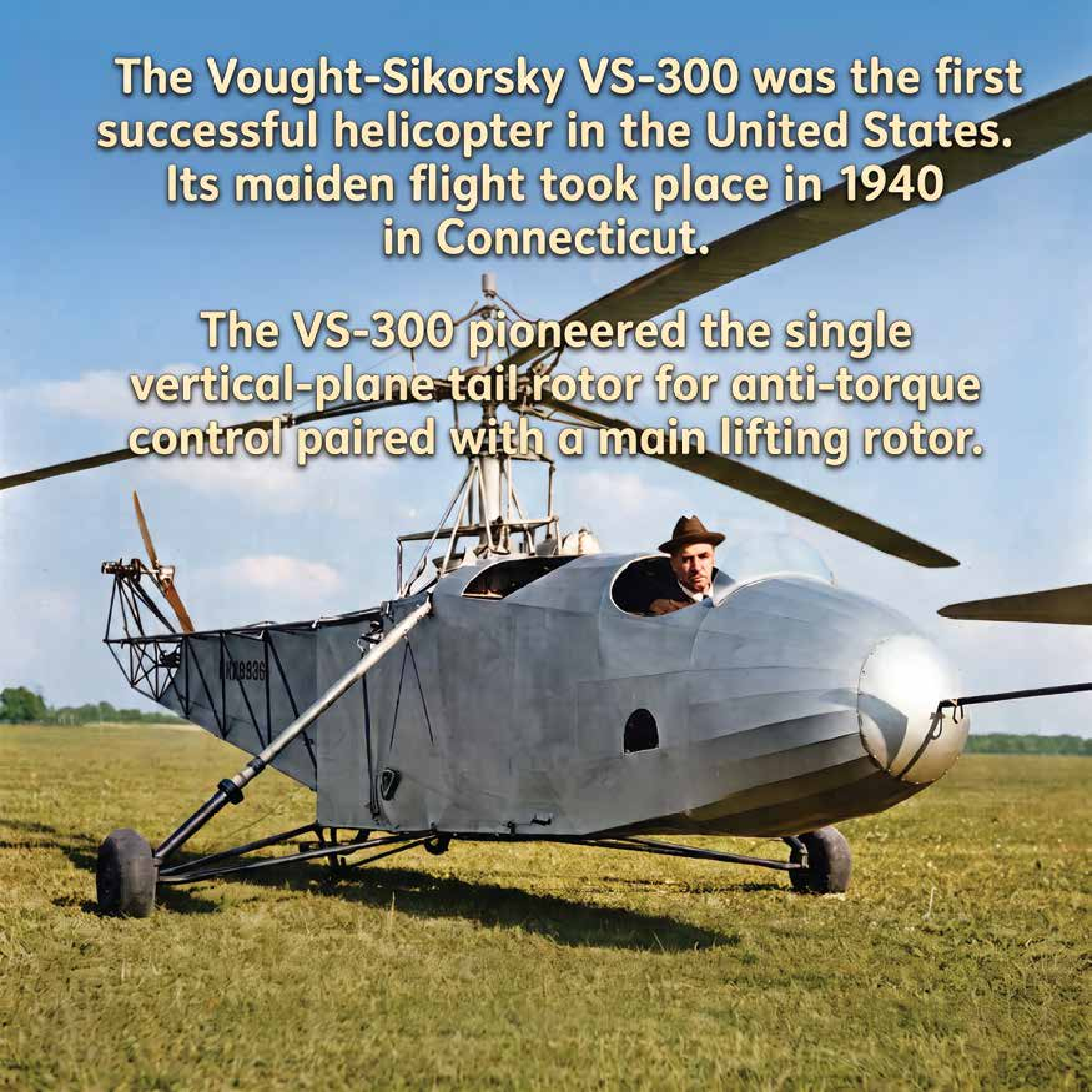
Vertical Take off and and Landing (VTOL)


N123HS

Leonardo da Vinci's aerial screw is the earliest known model for a helicopter. The screw was canvas over a wooden frame.



Unfortunately 15th century materials and engines were insufficient to build a working model. He had the right idea but not the technology he needed.


The Vought-Sikorsky VS-300 was the first successful helicopter in the United States. Its maiden flight took place in 1940 in Connecticut.
The VS-300 pioneered the single vertical-plane tail rotor for anti-torque control paired with a main lifting rotor.

The Bell 47 was introduced in 1946. It was the first helicopter in the world certified for civilian use.

The helicopter had a two-bladed main rotor with a stabilizer bar, a tail anti-torque rotor, twin external fuel tanks and simple tubular framework.

The Bell-Boeing V-22 Osprey was a mix of airplane and helicopter.

It could take off vertically. Then the propellers tilted to allow it to fly like a plane.

As electric batteries and computer controls have improved, it's become possible to build powerful drones.

In 2026, a farmer in Indonesia named Daeman commuted to work hanging by a rope from a CNA agricultural drone.


Jetson is building flying cars. These are single person electric aircraft.
The regulatory regime in the US makes flying them challenging. It’s possible other countries will use these devices before the US does.

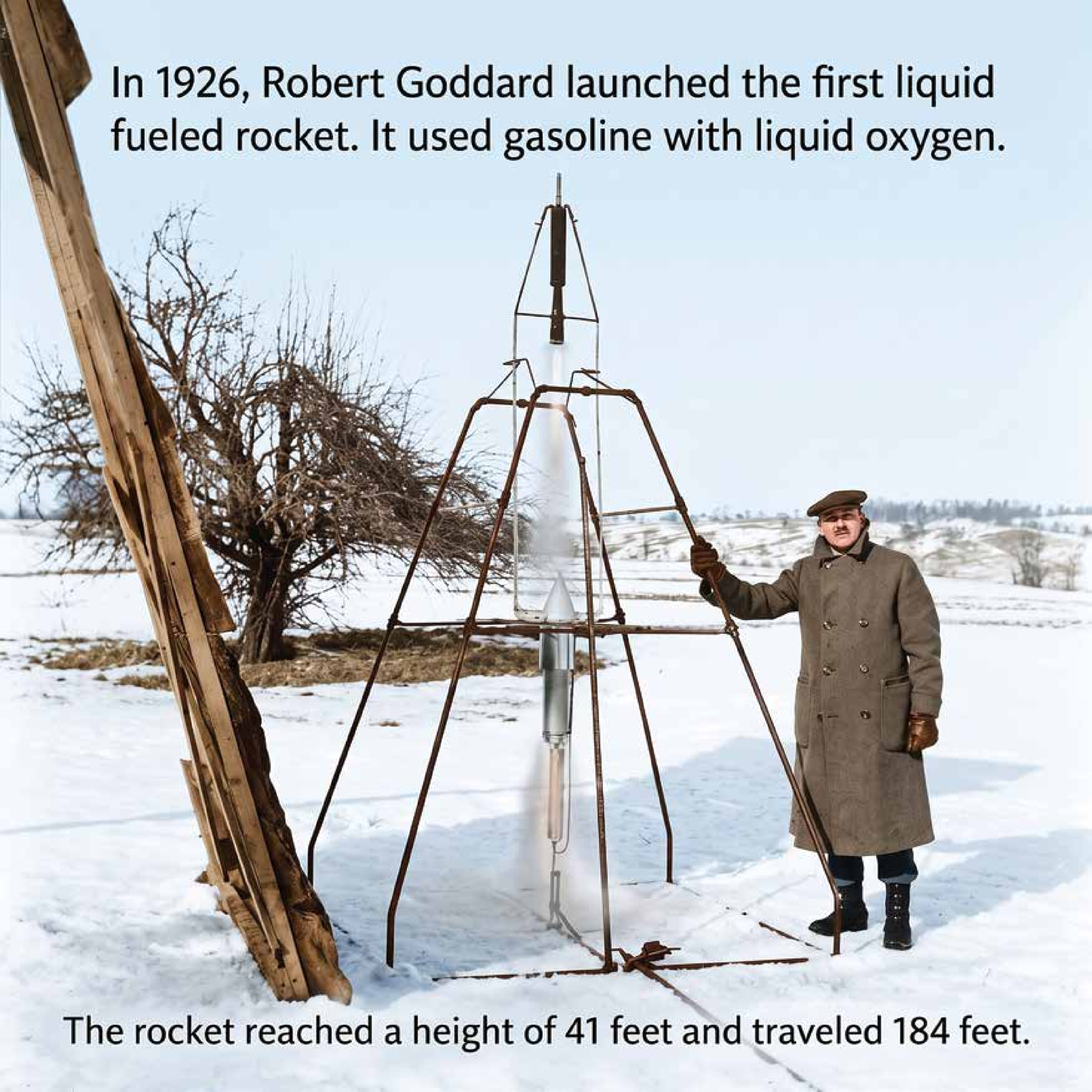

SPACE

During the Mongol siege of Kaifeng in 1232, defenders employed fire arrows.

These consisted of a paper tube packed with gunpowder attached to a standard arrow shaft. When ignited, gas escaping from the rear propelled the arrow forward under its own thrust.

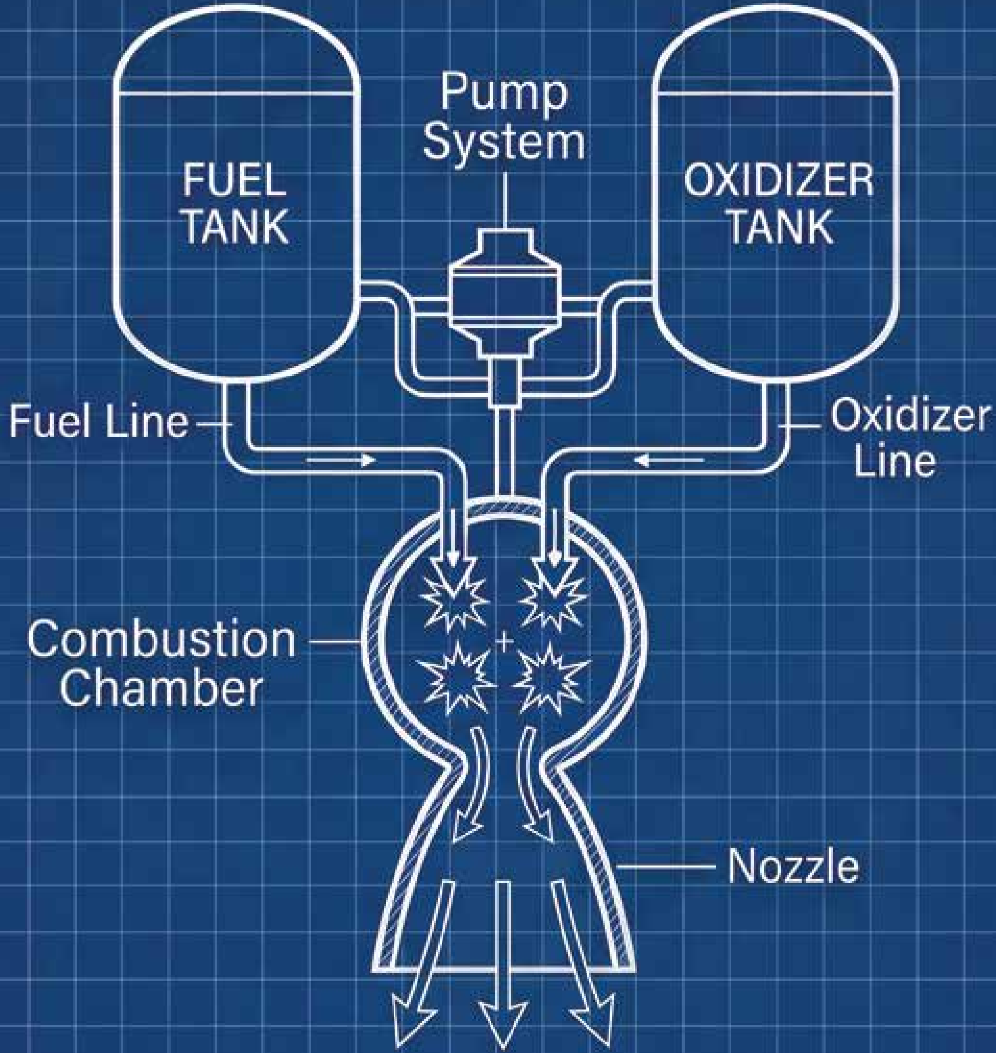

Over the following centuries, rockets spread across the world for military uses. In 1814, the British rocket ship HMS Erebus fired on US Fort McHenry.
This assault inspired Francis Scott Key to write the line "The rocket's red glare" in the Star Spangled Banner.

In 1926, Robert Goddard launched the first liquid fueled rocket. It used gasoline with liquid oxygen.



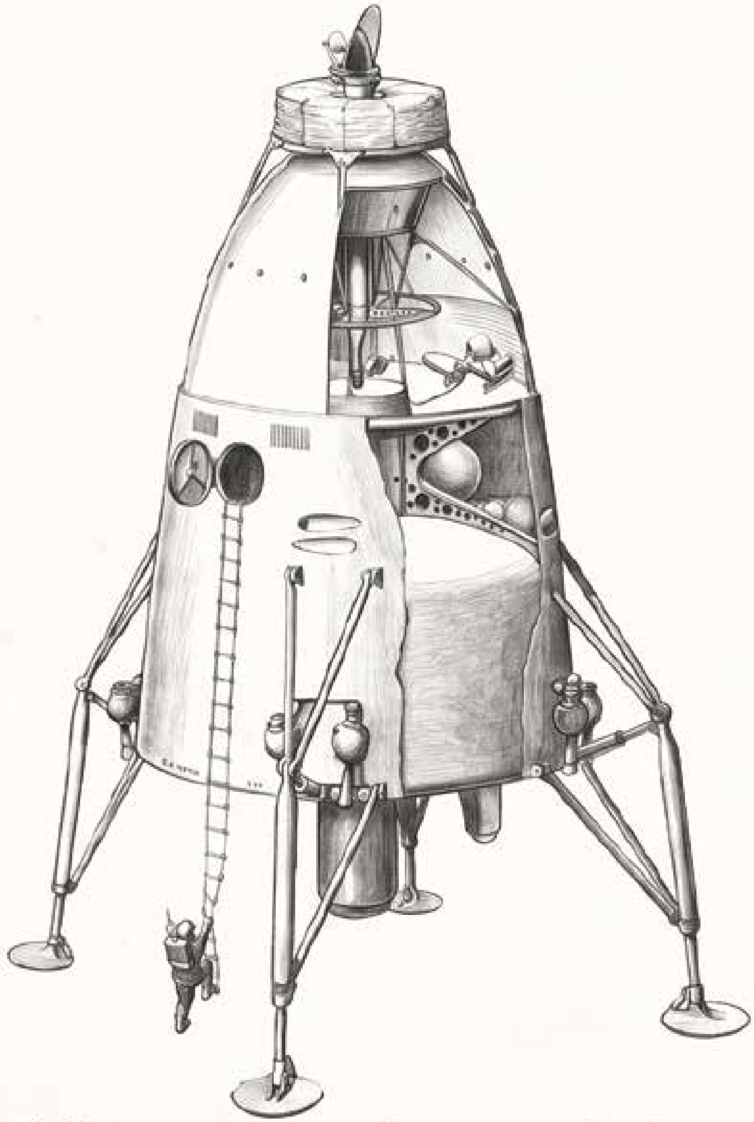
The rocket reached a height of 41 feet and traveled 184 feet.

A liquid rocket carries fuel and an oxidizer. A chemical reaction of the two creates thrust.



Goddard used gasoline and oxygen. Today hydrogen and oxygen are most common.

In 1934, Arthur C Clarke joined the British Interplanetary Society. Britain had outlawed explosive experiments in 1875. So, the society built mathematical models for rockets and satellites.



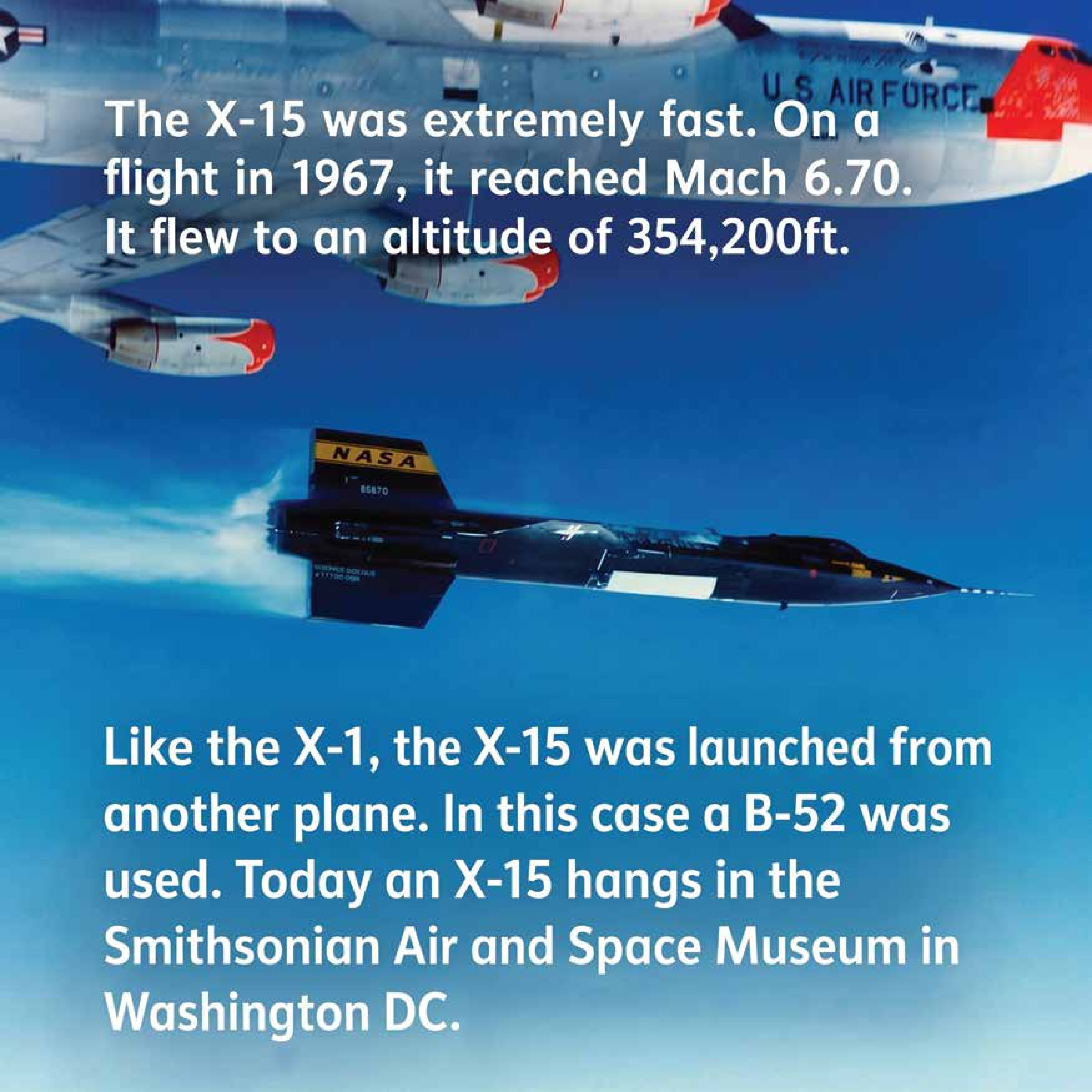
In the US and Germany work on rockets continued. If Clarke had been able to build and test rockets in the UK, it's quite possible they would have developed there first.

Werner von Braun was a visionary in rocket development. The Germans used his discoveries for weapons such as the V-1 and V-2 rockets.

After the war von Braun was recruited by the US. It seems possible that he was so fixated on building rockets he didn't particularly care who he worked for.

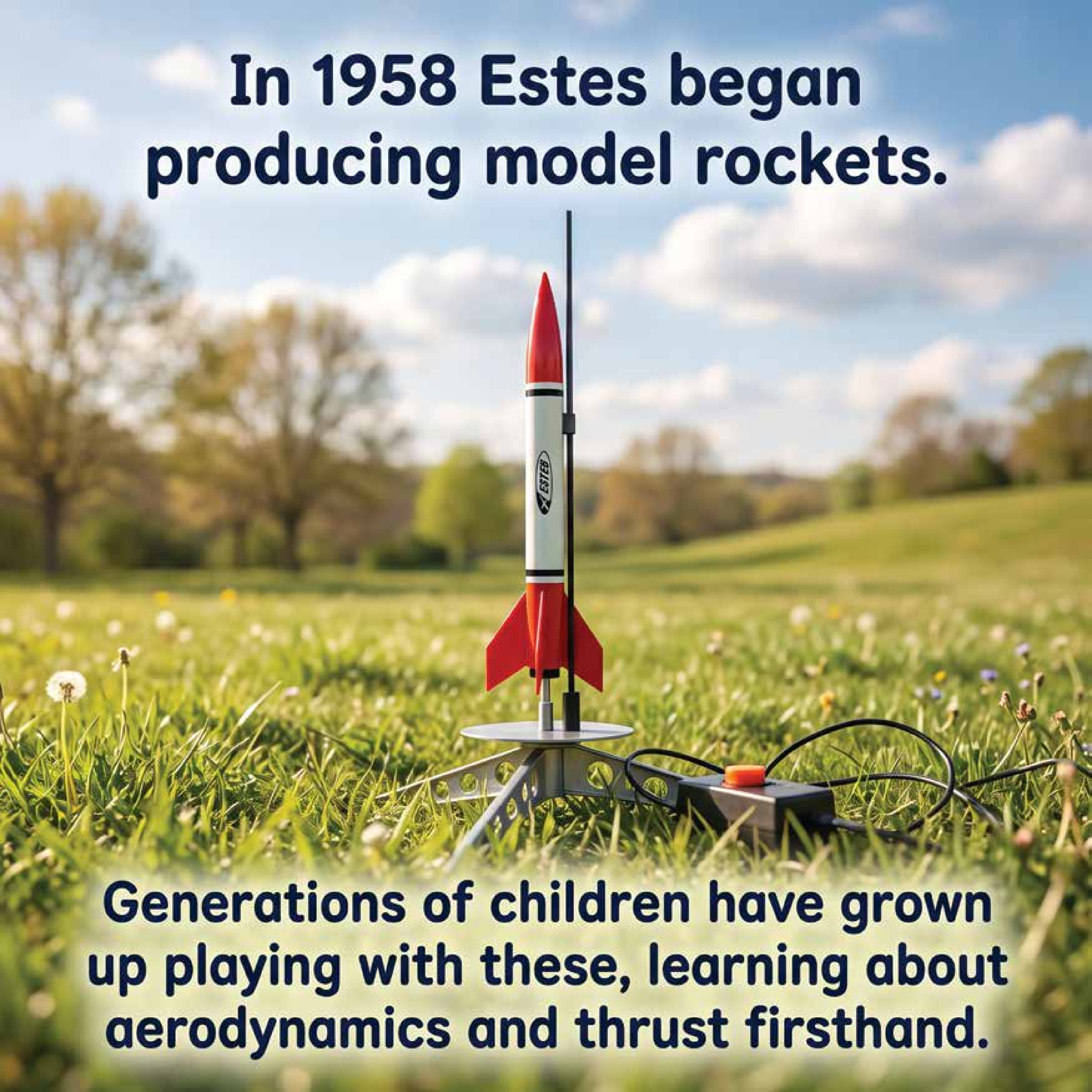
In 1947, Chuck Yeager became the first person to break the sound barrier.

The Bell X-1 was dropped from the bay of a B-29 Superfortress. In the first flight, Yeager piloted it to Mach 1.06



The X-15 was extremely fast. On a flight in 1967, it reached Mach 6.70. It flew to an altitude of 354,200ft.

Like the X-1, the X-15 was launched from another plane. In this case a B-52 was used. Today an X-15 hangs in the Smithsonian Air and Space Museum in Washington DC.


In 1958 Estes began producing model rockets.
ESTES
Generations of children have grown up playing with these, learning about aerodynamics and thrust firsthand.


In 1957, the Russians launched Sputnik. This was the first time a man made object achieved orbit.
Sputnik broadcast a simple signal for 21 days before its battery died. After 92 days the orbit deteriorated and Sputnik burned up on reentry.

Yuri Gagarin was the first person to journey into space and orbit the earth.

In 1961, the Vostok 1 carried Gagarin to an altitude of 203 miles. His flight lasted 108 minutes. He completed a single orbit around the Earth at 17,000 mph.


In 1962 president John F Kennedy spoke at Rice. America was behind in the space race. Kennedy wanted to fix that.
SEAL OF THE PRESIDENT OF THE UNITED STATES
He said, "We choose to go to the moon. We choose to go to the moon in this decade and do the other things, not because they are easy, but because they are hard."

The Apollo 11 mission put a man on the moon in 1969. Neil Armstrong, Buzz Aldrin and Michael Collins launched on top of a Saturn V rocket.

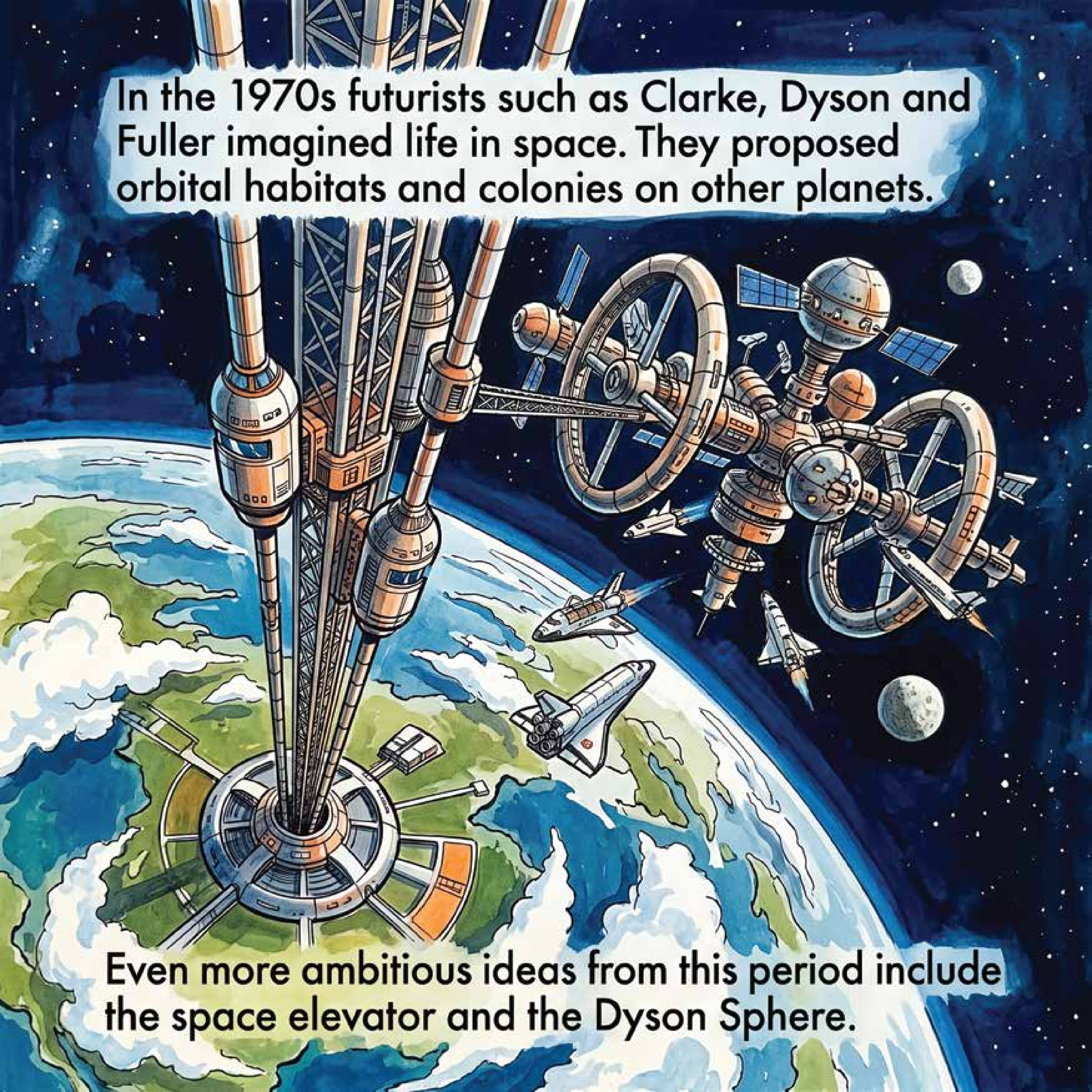
The rocket had 3 stages. As it flew higher, those fell off. A lander and orbiter then made the journey to the moon.

After 3 days, the orbiter and lander arrived at the moon. They separated. The lander descended to the surface of the moon.

After it landed, Neil Armstrong said, "Houston, Tranquility Base here, the Eagle has landed."

Neil Armstrong was the first man to set foot on the moon.

He said, “That’s one small step for man, one giant leap for mankind.”


In the 1970s futurists such as Clarke, Dyson and Fuller imagined life in space. They proposed orbital habitats and colonies on other planets.
Even more ambitious ideas from this period include the space elevator and the Dyson Sphere.

First launched in 1981, the Space Shuttle was intended to be a reusable space craft.

It launched on 3 rockets. Descending from space, it flew like a plane, ultimately landing on a runway.

The Space Shuttle had numerous challenges. The heat shield tiles failed, requiring expensive inspection and replacement after each flight.

The rockets were extremely complex, leading to the Challenger explosion in 1986. The Columbia exploded in 2003. The program was abandoned.

In 2003, Burt Rutan flew SpaceShip One. This was the first private spacecraft.

It had two parts. A plane called White Knight flew to 50,000 ft. SpaceShipOne detached and traveled to a height of 70 miles.

In 2002, Elon Musk founded SpaceX. It supplied rockets to NASA. SpaceX has launched over 11,000 Starlink satellites. They form a mesh network that provides internet access all over the world.

Communication satellites were orignally proposed by Clarke in 1945. He suggested a geosynchronous orbit, not predicting mesh networks and computer advancements.

The ultimate goal of SpaceX is colonizing Mars. Starship is intended to get us there.

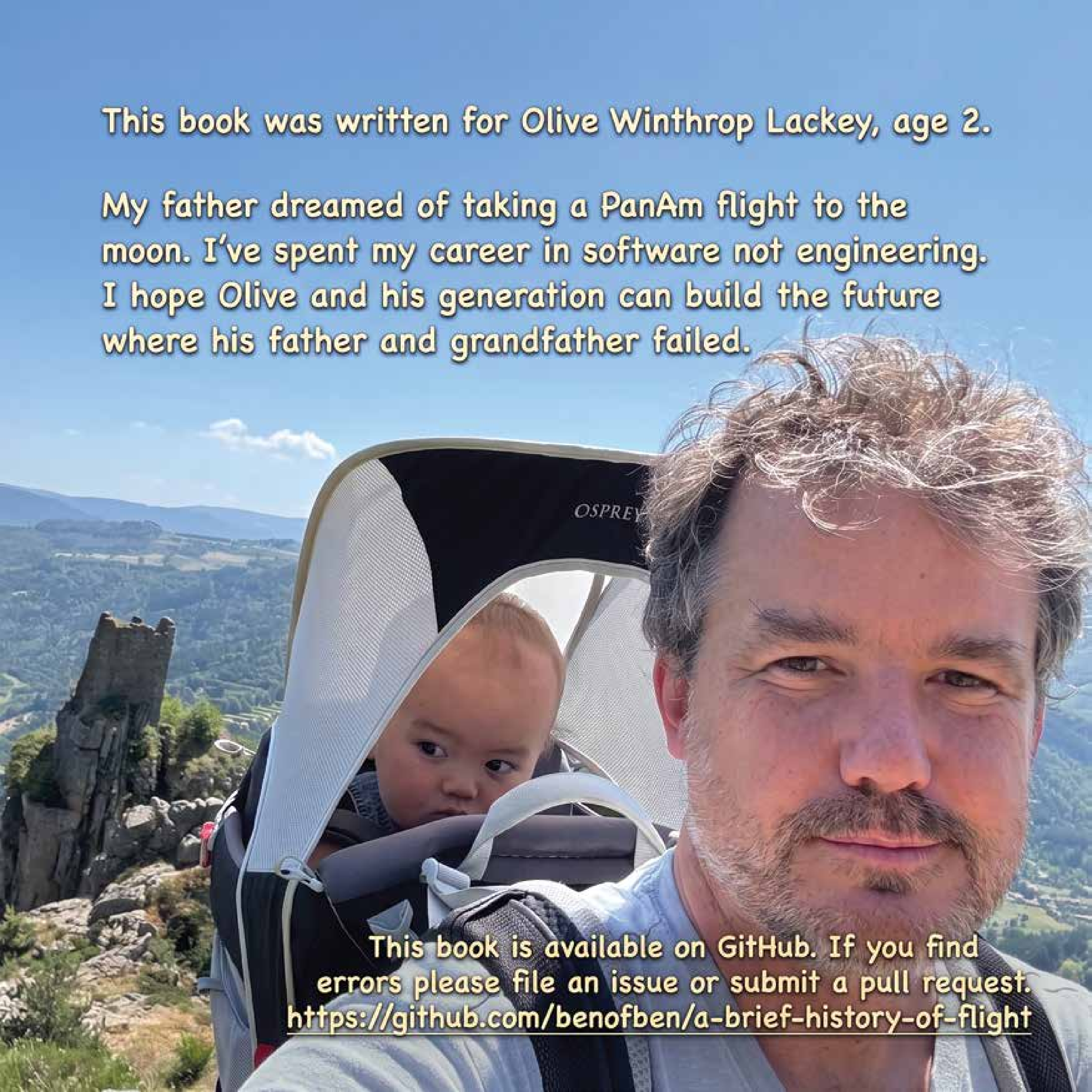
It flys out of Boca Chica at the southernmost point in Texas. Orbital insertion requires less energy closer to the equator.

SpaceX is planning unmanned missions to Mars with Optimus robots. Those will be followed by manned missions.

It is quite possible that you will be able to travel to Mars. That will take work, engineering and perserverance.

This book was written for Olive Winthrop Lackey, age 2.

My father dreamed of taking a PanAm flight to the moon. I've spent my career in software not engineering. I hope Olive and his generation can build the future where his father and grandfather failed.



This book is available on GitHub. If you find errors please file an issue or submit a pull request.
https://github.com/benofben/a-brief-history-of-flight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